Generalizing antibodies' developability predictions across studies

Project proposed by the Gray lab, Johns Hopkins university

Antibodies are vital proteins in therapeutics, but predicting their properties across datasets poses ML challenges. This brief outlines the hackathon task using FLAb to tackle cross-study generalization.

Important notes:

- *An online meeting is set up for you with Michael Chungyoun from the Gray lab on Saturday 8pm. It's a great opportunity to ask your questions.*
- *This challenge is quite different from the other ones in that it does not require (in theory) to use the major protein design models we make available (RFD3, BoltzGen).*

Introduction and Background

Antibodies are Y-shaped proteins produced by all animals with an adaptive immune system. Plasma B cells produce them by undergoing clonal selection and affinity maturation, creating high-affinity variants specifically tailored to neutralize an antigen, a specific molecule on the invading virus or bacteria. The antigen can be a protein, a lipid, a nucleic acid, or other molecules. The specific molecular pattern recognized and bound on the antigen is called the epitope. Key structural features of antibodies include two heavy and two light chains. These chains form the Fab regions, the antibody's arms, whose ends bind the antigen through hypervariable complementarity-determining regions (CDRs). The H3 CDR loop, found on the heavy chain, is significantly more variable, often contributing the most contacts and antigen-binding energy.

Monoclonal antibodies (mAbs) are being developed in the lab by the pharmaceutical industry to treat a variety of illnesses such as cancer, Alzheimer's and asthma. A single B cell is cloned and its lineage produces identical antibodies that target a specific epitope. mAbs are one of the most promising avenues in pharma with a market size of between USD 200 billion and USD 300 billion in 2025. They accounted for 25% of drugs newly approved by the FDA in 2025. *Other than epitope specificity, mAbs need to showcase desirable developability properties to become a drug including high thermostability (Tm >70°C), low immunogenicity, high expression yield, low aggregation, etc.* These often involve trade-offs, where enhancing a property can worsen another.[1]

Machine-Learning Approaches for Antibodies

Many protein structure and property prediction models rely on co-evolutionary patterns in multiple sequence alignments (MSA). Amino acids that have evolved together over the course of history tend to be close in 3D space. Additionally, sequences with high similarity across species tend to occupy the same function. Because antibodies are not inherited across evolution and evolve independently within each organism, MSAs are not always available for antibodies.[1]

For this reason, certain models are trained specifically on antibodies for structure prediction (IgFold) and inverse folding (AbMPNN). *Notably, protein language models use a transformer architecture similar to the one powering today's renowned chatbots (Gemini, ChatGPT). They create embeddings and perplexity [8] measures, both of which can be used for prediction of developability properties.* Among these, AntiBERTy [2] and IgLM [3] are trained specifically on antibodies while the ESM family [4] and ProGen [5] includes a diversity of different proteins.

The FLAb Dataset

"The Fitness Landscape for Antibodies (FLAb) [6] is the largest publicly available therapeutic antibody dataset designed to train and benchmark protein AI models. [...] FLAb encompasses over 3 million antibody developability assay data points, aggregated from public studies." Flab was developed by Michael Chungyoun from the Gray Lab at Johns Hopkins University, who you will meet Saturday night, in collaboration with AWS. Its main goal is to benchmark models that predict developability properties for antibodies.

The Generalization Problem

Models trained on one study's data can yield acceptable results in-study. In other words, they achieve similar accuracy between train and test sets when both sets are taken from the same experimental conditions. However, evaluating such a model on a held-out study set typically causes a significant drop in accuracy. This happens because inter-study differences like experimental protocols and expression hosts cause shifts in the distribution of developability properties that the sequence models do not capture.

As in-vitro methods to test developability properties are expensive, pharmaceutical companies are increasingly relying on computational methods for screening potential candidates. However, the generalization problem means a model trained on a set of studies cannot provide a reliable screening criteria for a new therapeutic target, expression host or set-up, seriously limiting the usefulness of such methods.

Your Task

In this project, you are asked to complete the following tasks:

- **Choose one or more developability properties.** Your choice can rely on the amount of data available in the FLAb dataset, the success of existing methods for prediction, etc.Thermostability, expression or polyreactivity are examples of such interesting developability properties *(Michael's recommendation)*. Stay away from binding affinity as the diversity of targets in the dataset will complicate the task.
- **Don't take our word for it!** Demonstrate the generalization problem yourself. Train a model that predicts developability properties. Evaluate it on both an in-study test set and a held-out study set. Compare performances. You can inspire yourself from FLAb's few-shot methods [6] (e.g. using pre-trained models' embeddings + MLP)
- **Try to solve the generalization problem.** This is the core of the challenge :) You can inspire yourself from already existing methods for shift correction like DomainBed [7] or come up with your own.

Success is defined as finding a method that produces **better held-out study set metrics than the baseline you'll have defined in the second task.** If time permits, you can even create your own HuggingFace space in the hugging-science organization to showcase your model!

Pitch presentation

Convince the jury that your project is the best!
You will present a **5 minute pitch** on Sunday, 15:00. You are encouraged to use a power-point presentation; send it to dlai@epfl.ch in PDF format **before Sunday 14:45!**

- **Explain your problem and why it is important.**
- **What was your computation workflow?**
- **What are your in-silico results?**
- **What are the project's limitations?**
- **How would you test it in the lab?**

Good luck and Enjoy the Ride!

Figures

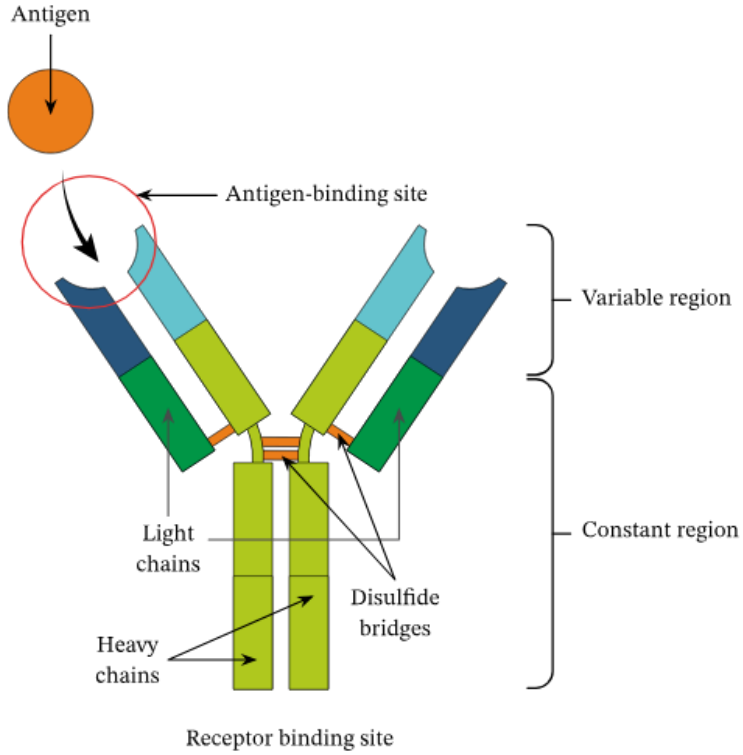

Figure 1: Schematic representation of an antibody structure.

References

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